

PART A: OBJECTIVE QUESTIONS (20 MARKS)

1. Which of the following statements is the specification of non-ideal pipelining?
 - A. It has just two segments and only one process can be run in a time.
 - B. It has several segments and all the segments have the same execution time.
 - C. It consists of several segments with various execution times.
 - D. The interface delay for the pipelining is very short.

2. Which of the following statements is NOT a limitation in pipelining architecture?
 - A. Resource Conflict
 - B. Data Dependency
 - C. Execution Cycle
 - D. Branch Difficulties

3. Which of the following is NOT one of the control unit inputs?
 - A. Instruction register
 - B. Control signal from control bus
 - C. Control signal to memory
 - D. Flags

4. Micro-operations include the data transferring
 - A. Into or out of a register
 - B. Between register to register
 - C. Between register to external bus
 - D. All of the above.

5. Which of the following facts about LOI or HOI is **FALSE**?
 - A. In high-order interleaving (HOI), the high order bits specify the memory bank.
 - B. In LOI, a failure of any single bank would be catastrophic to the whole system.
 - C. LOI provides better reliability, since a failed module affects only a localized area of the address space.
 - D. Easier to extend memory modules in HOI.

6. In Set Associate Mapping, read or write a word at the memory address 357A is equal to 0011 0101 0111 1010. This address will be translated to _____.
 - A. tag = 001101, slot=0101, word = 1111010
 - B. tag = 001101, line = 01011, word = 11010
 - C. set=0011, tag = 010101, word = 1111010
 - D. tag = 001101, set= 010111, word = 1010

7. Computer memory is organized in a hierarchy, with the _____i_____, _____ii_____ memory at the top and the largest, _____iii_____ memory at the bottom.
- A. i=largest, ii=fastest, iii=slowest
 B. i=largest, ii=slowest, iii=fastest
 C. i=smallest, ii=fastest, iii=slowest
 D. i=smallest, ii=slowest, iii=fastest
8. _____ enhances performance by providing greater memory capacity, without the expense of adding main memory. While, _____ memory enhances performance by providing faster memory access speed.
- A. replacement algorithm, virtual
 B. Virtual memory, cache
 C. Cache memory, virtual
 D. Dynamic memory, static
9. The performance of a replacement algorithm for cache also known as the Average Memory Access Time (AMAT), can be computed using _____.
- A. $AMAT = \text{Hit rate} \times \text{Hit time} + \text{Miss penalty}$
 B. $AMAT = \text{Hit rate} + (\text{Miss rate} + \text{Miss penalty})$
 C. $AMAT = \text{Hit time} + (\text{Miss rate} \times \text{Miss penalty})$
 D. $AMAT = \text{Hit ratio} + (\text{Miss rate} \times \text{Miss cycle})$
10. Which of the following is NOT a measure for computer performance?
- A. Response time
 B. RAM size
 C. Throughput
 D. Execution time
11. Performance of computer can be measured based on the following **EXCEPT**?
- A. Speed
 B. Throughput
 C. Read Time
 D. Response Time
12. To achieve better performance, RISC machines reduce _____, meanwhile CISC reduce _____.
- A. the number of cycles per instruction, the number of instructions per program
 B. the number of instructions per program, the number of cycles per instruction
 C. the program execution time, the clock cycle
 D. the clock cycle, the program execution time

13. Which of these is a type of I/O devices?
- A. Integer I/O devices
B. Block I/O devices
C. Bit I/O devices
D. Stream I/O devices
14. The input output subsystems include
- I. Blocks of main memory
II. Programmed polling
III. Control module
IV. Interfaces
V. Cabling
- A. All of the above
B. I, II, III and IV
C. I, II, IV and V
D. I, III and V
15. The input output can be controlled using the input output channel. The I/O processors (IOPs) serves as
- I. Negotiates protocols
II. Issues device commands
III. Translates storage coding to memory coding
IV. Transfer entire files or groups of files independent of the host CPU.
- A. All of the above
B. II, III and IV
C. I, II and IV
D. III and IV
16. If **AX = 3254H** and **BX = 2B2CH**, what is the result of **AND AX, BX**?
- A. AX = 2202H
B. AX = 4404H
C. AX = 2204H
D. AX = 4402H
17. Study the assembly code given. What is the final result of register CH?

MOV AH, 3
MOV BH, 14
MOV CH, 7
ADD AH, BH

```
JNZ HERE  
  
MOV CH, 1  
  
JMP QUIT  
  
HERE:  
  
MOV CH, 5  
  
QUIT:
```

- A. 1
- B. 5

- C. 7
- D. 0

18. Based on the assembly code below. Please identify if the line indicated is a valid code.

```
.DATA  
  
SUM WORD 126  
  
TOTAL WORD 55H  
  
  
.CODE  
  
MAIN PROC  
  
MOV AX, BH ;LINE1  
  
ADD SUM, TOTAL ;LINE2  
  
SUB AX, SUM ;LINE3
```

- A. LINE1 = valid; LINE2 = invalid; LINE3 = valid
- B. LINE1 = invalid; LINE2 = invalid; LINE3 = valid
- C. LINE1 = invalid; LINE2 = invalid; LINE3 = invalid
- D. LINE1 = valid; LINE2 = valid; LINE3 = valid

19. Study the assembly code given below. If the first user input is 12 and the second is 8, what is the correct output?

```
.DATA

STR1 BYTE "ENTER A NUMBER :",0

.CODE

MAIN PROC

MOV EDX, OFFSET STR1

CALL WRITESTRING

CALL READDEC

MOV EDX, OFFSET STR1

CALL WRITESTRING

CALL READDEC

CALL WRITEDEC

CALL WRITEDEC
```

- A. 12 8
B. 12 12
C. 8 12
D. 8 8
20. Study the assembly code given below. Choose the correct output.

```
.DATA

STR1 BYTE "Test1 ",0

STR2 BYTE "Test2 ",0

TREST BYTE "aBc ", 0

STR3 BYTE "Test3 ",0


.CODE

MAIN PROC

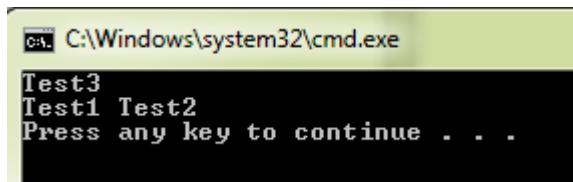
MOV EDX,OFFSET TREST
```

```
MOV EAX, OFFSET STR3  
CALL WRITESTRING  
CALL CRLF
```

```
MOV EDX, OFFSET STR1  
CALL WRITESTRING
```

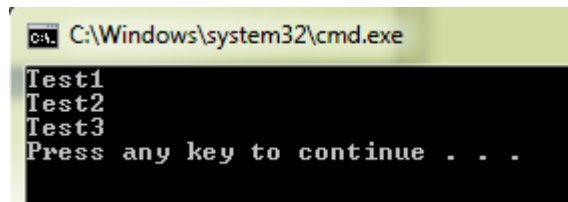
```
MOV EDX, OFFSET STR2  
CALL WRITESTRING  
CALL CRLF
```

A.



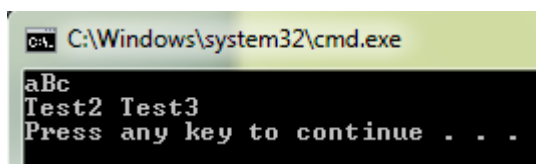
C:\Windows\system32\cmd.exe
Test3
Test1 Test2
Press any key to continue . . .

C.



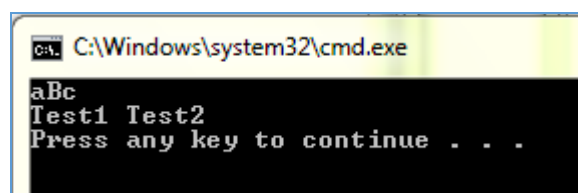
C:\Windows\system32\cmd.exe
Test1
Test2
Test3
Press any key to continue . . .

B.



C:\Windows\system32\cmd.exe
aBc
Test2 Test3
Press any key to continue . . .

D.



C:\Windows\system32\cmd.exe
aBc
Test1 Test2
Press any key to continue . . .

PART B: STRUCTURED QUESTIONS (80 MARKS)

QUESTION 1 (15 MARKS)

- (a) A pipeline processor has 6 segments - *Fetch Instruction* (FI), *Decode Instruction* (DI), *Calculate Operand* (CO), *Fetch Operand* (FO), *Execute Instruction* (EI) and *Store Operand* (SO). There are five instructions (I1 – I5) involved, they are:

I1: CMP AH, AL
I2: MUL CH, 2
I3: JMP I5
I4: ADD DH, 011F
I5: DIV BX, 3

- i. Draw the space time diagram for the above instructions. [3]
 - ii. Solve the branching problem using NOP. [4]
- (b) Table 1 shows the delay time for four segments in a pipelining architecture. If the interface delay is 10 nanoseconds, calculate:

Table 1: Delay time for segments in pipelining.

Segment Name	Delay Time (nanoseconds)
Segment 1	40
Segment 2	25
Segment 3	30
Segment 4	30

- i. The cycle time of the pipeline, t_p . [2]
- ii. The execution time for 200 tasks. [4]
- iii. The real and the maximum speed up for 200 tasks. [2]

QUESTION 2 (5 MARKS)

A system uses a control memory of 4096 words of 46 bits each. The microinstruction has three fields, i.e., select, address and microoperations fields.

- (a) If the microoperation field requires 24 bits, how many bits are there in the branch address field and the select field? [2]
- (b) If there are 64 status in the system, how many bits of the branch logic are used to select a status? [2]
- (c) How many bits are left to select an input for the multiplexers? [1]

QUESTION 3 (20 MARKS)

- (a) Differentiate between SRAM and DRAM. [2]
- (b) Based on the following specifications: [6]
 - i. 24-bit memory address, each address can hold 24 bits. What is the memory capacity?
 - ii. 32 Mbytes memory divided into 64-kilobit blocks. How many blocks available?
 - iii. 2 Mbytes memory is divided into several pages. There are 128 pages. What is the size of each page?
- (c) Given a memory address as 34Ah (10 bits) and there are 4 memory banks. Determine the memory bank address and the address of the word in the bank using LOI. [4]
- (d) A computer main memory is utilizing a Block Direct Mapping:
 - i. Given the main memory address format as below, what is the total memory capacity, if each main memory word is 8 bits? [3]

5	8	3
Tag	Line	Word

- ii. The total numbers of cache words? [2]
- iii. Suppose a program generates an address 246Ah and then the program generates the address 256Ah, will it be a hit or a miss? Show your working. [3]

QUESTION 4 (10 MARKS)

(a) Identify the correct terms for **A to E**. Select from answer options below. [5]

Description	Term
It consists one or more I/O processors (IOPs) that control various channel paths.	i
It less wasteful of CPU cycles than programmed I/O.	ii
It has direct access to the memory (steals bus cycles from the CPU).	iii
It delegates I/O to intelligent I/O device controllers called I/P processors that can execute programs with branching instructions.	iv
Two dominant recording methods are <i>serpentine</i> and <i>helical scan</i> , which are distinguished by how the read-write head passes over the recording medium.	v

Answer options:

Channel I/O	Interrupt Driven I/O	Programmed I/O
Direct Memory Address	Magnetic Tape	Optical Disks

(b) The following parameters are used for processor overhead calculation for interrupt I/O and DMA. [5]

- 500 MHz CPU
- Interrupt handler takes 400 cycles
- Data transfer takes 100 cycles
- 4 MB/s, 16 B interface disk transfers data 50% of time
- DMA setup takes 800 cycles, transfer one 16KB page at a time

- i. Calculate processor overhead for interrupt I/O.
- ii. Calculate processor overhead for DMA.

QUESTION 5 (20 MARKS)

(a) Given the pseudocode below, find the arithmetic expression for it. [4]

```
MOV AX, SUMA
MOV BL, SUMB
DIV BL
INC AL
MOV TOTAL, AL
```

(b) Given the pseudocode below, find the content of registers as indicated. [3]

```
MOV CL, BFh
MOV BL, 10h
NOT CL      ;CL =   i  
AND BL, CL ;BL =   ii 
OR  BL, 5  ;BL =   iii 
```

- (c) Given the array definition below, write the complete assembly code to sum the content of array that in odd position by using *loop*. Then, print the value of sum (in decimal) as an output. [7]

```
.data
Array2 word 25h, 1A20h, 66h, 891h, Ah
```

- (d) Given the pseudocode below, write the appropriate assembly code to achieve it. [6]

```
If BX < AX AND Y = 0 THEN
    Y = Y - 1
ELSE
    Y = BX + Y
```

QUESTION 6 (10 MARKS)

- (a) Table 2 depicts the CPU Execution Time of Computer A and B.

Table 2: CPU Execution Time for Computer A and B

	Clock cycle	CPI	Clock rate
Computer X	1000ps	3.0	4GHz
Computer Y	1500ps	2.0	4GHz

Note: The formula

CPU time = Instruction count x CPI x Clock cycle time

Or

CPU time = $\frac{\text{Instruction count} \times \text{CPI}}{\text{Clock rate}}$

- i. Calculate the CPU time for each computer. [4]

$$\begin{aligned} X &= 1000 * 3.0 * 4 * 10^9 \\ &= 12000 * 10^9 \\ &= 1.2 * 10^{9+4} \\ &= 1.2 * 10^{13} \end{aligned}$$

$$\begin{aligned} Y &= 1500 * 3.0 * 4 * 10^9 \\ &= 18000 * 10^9 \\ &= 1.8 * 10^{9+4} \\ &= 1.8 * 10^{13} \end{aligned}$$

- ii. From the CPU time calculated, which computer should be selected based on the performance? Discuss your observation. [3]

- (b) Explain the importance of benchmark in CPU design. [3]